

Boltz Ewok Setup

Table of Contents

Table of Contents	1
1. Prerequisites	2
1.1 Install VS Code.....	2
1.2 Github Account	2
1.3 Remote - SSH extension	2
2. Connect to the Ewok Machine	3
3. Clone the Boltz Repository.....	4
4. Set Up the Boltz Environment.....	5
4.1 Install Conda.....	5
4.2 Run the Boltz Installer	7
4.3 Run the Boltz UI Notebook	7

1. Prerequisites

1.1 Install VS Code

Download and install VS Code from the official download page:

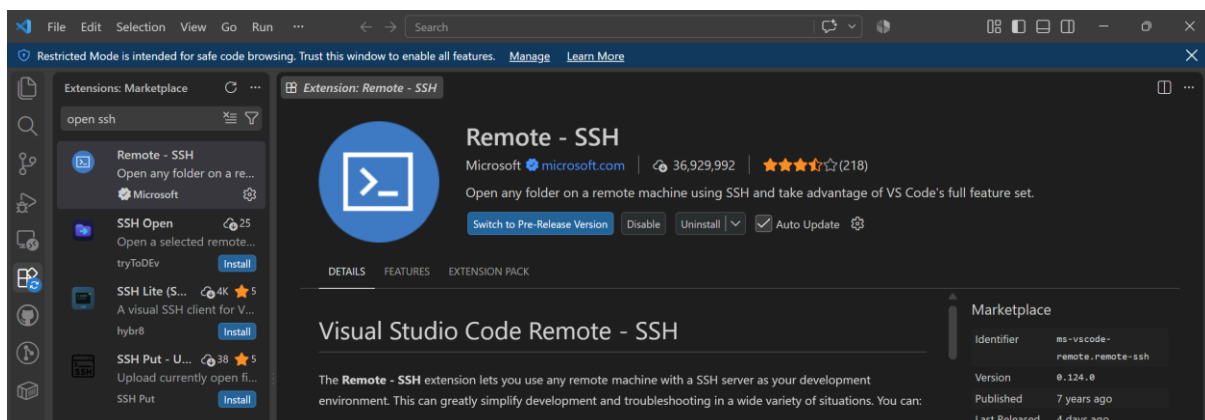
https://code.visualstudio.com/download?exp_download=fb315fc982

1.2 Github Account

Create a GitHub account at <https://github.com/> if you do not already have one.

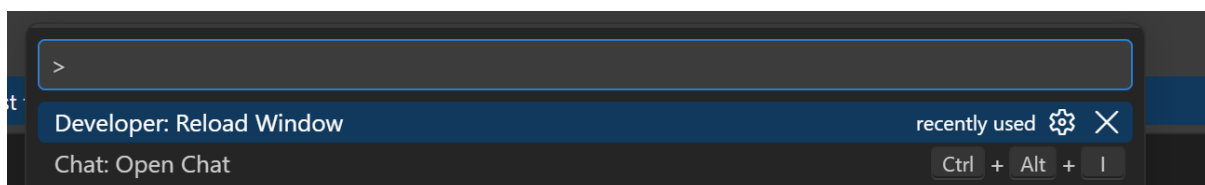
1.3 Remote- SSH extension

Install the **Remote – SSH** extension in VS Code. This extension enables you to connect to an Ewok machine from your local computer and access its computational resources directly through VS Code.



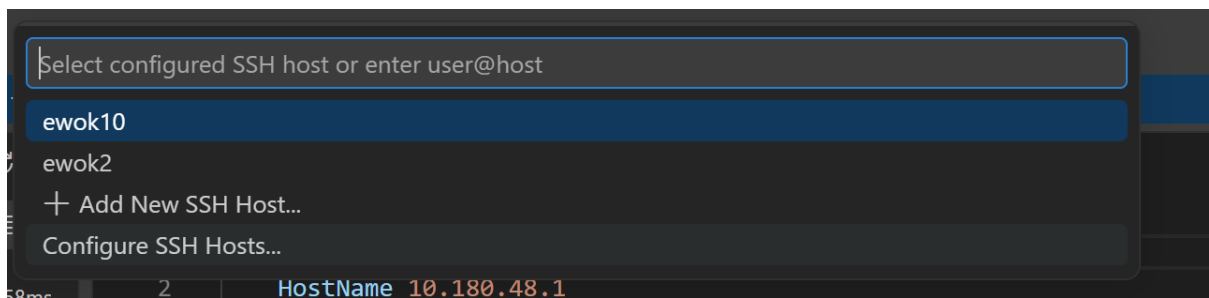
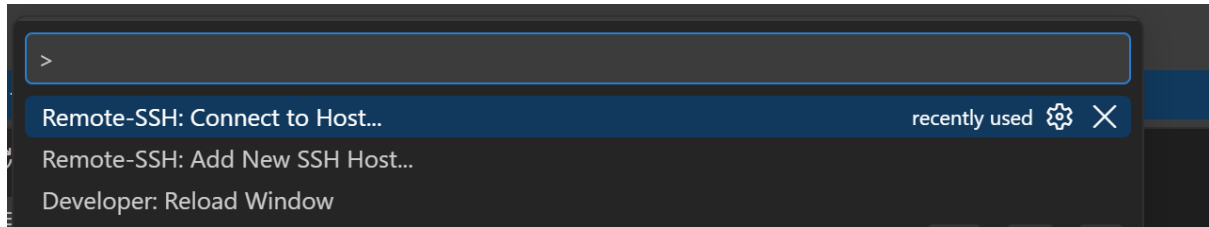
After installation, open the **Command Palette** (**Ctrl + Shift + P** on Windows or **Cmd + Shift + P** on Mac.) and type **>Developer: Reload Window**, and press Enter to ensure the extension loads correctly.

Quick tip: To run any command in the **Command Palette**, make sure to start it with a **>** symbol, e.g. **>your command**.



2. Connect to the Ewok Machine

In VS Code, open the **Command Palette** and type **>Remote-SSH: Connect to Host**, then select **Configure SSH Hosts**.



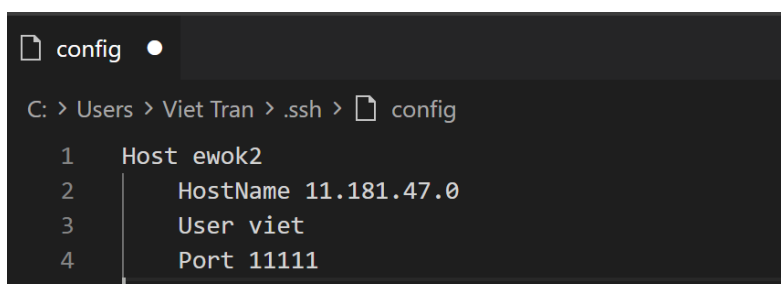
Add an entry for your Ewok machine using the following configuration format:

```
Host <ewok_machine_number>
  HostName <your_ewok_machine_ip_address>
  User <your_ewok_username>
  Port <your_ewok_port>
```

Example configuration:

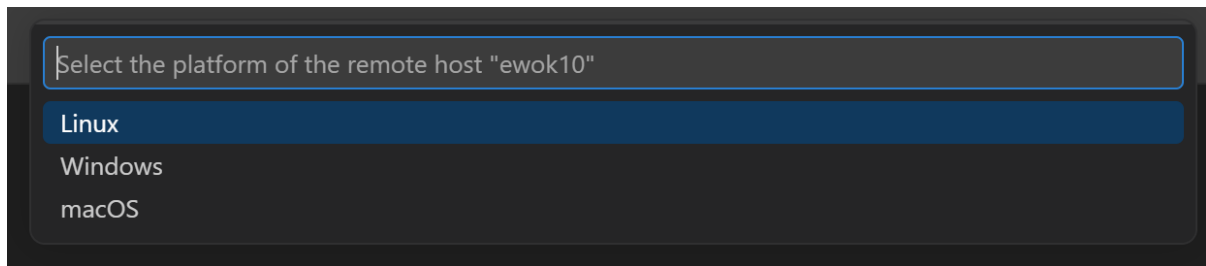
```
Host ewok2
  HostName 11.181.47.0
  User viet
  Port 11111
```

Make sure to save your changes to the config via **Ctrl + S** on Windows or **Cmd + S** on Mac.

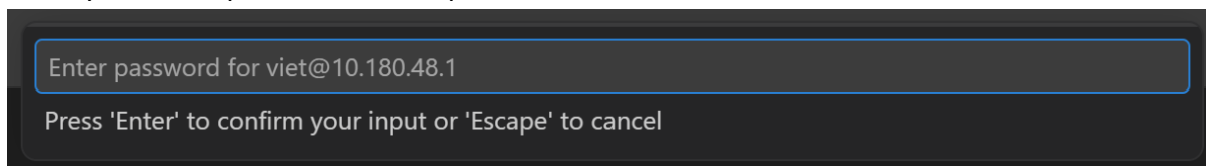


Note: Contact [Zhen He](#) or [Aaron Harris](#) for setting up and access your Ewok account.

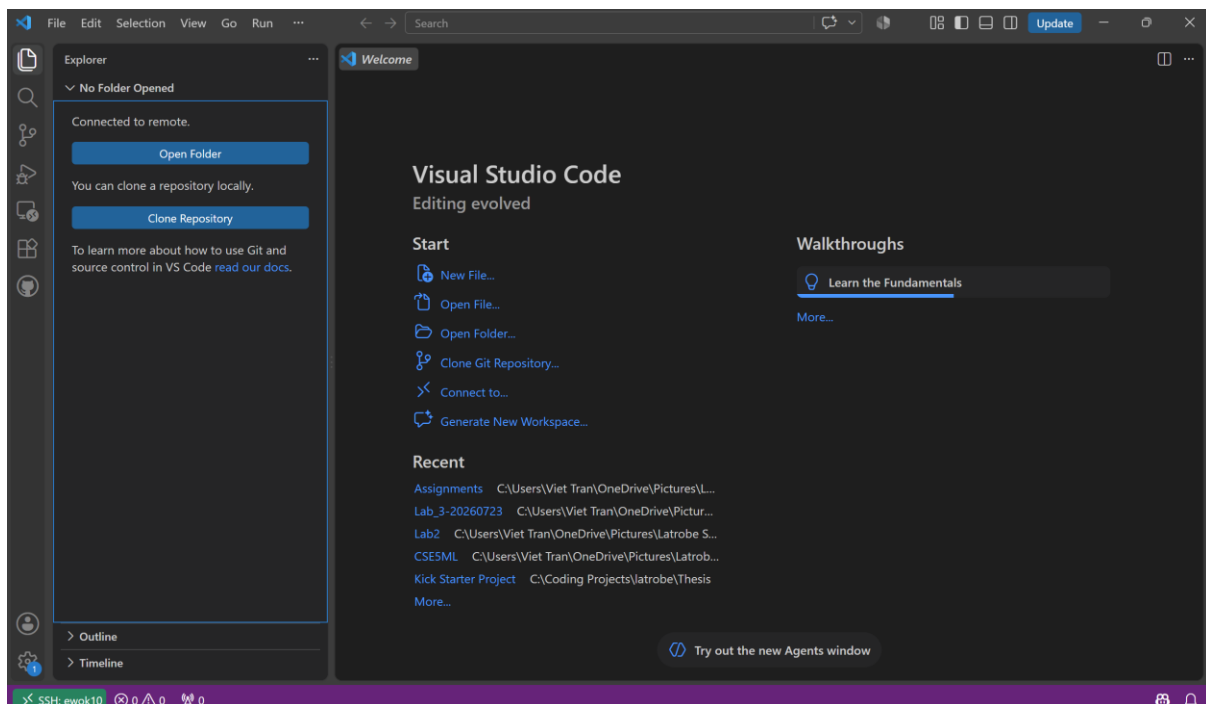
If VS Code prompts you to select the remote host's operating system, choose **Linux**



Enter your Ewok password to complete the connection.



Once the connection is established, the Ewok host name should appear in the bottom-left corner of the VS Code window.

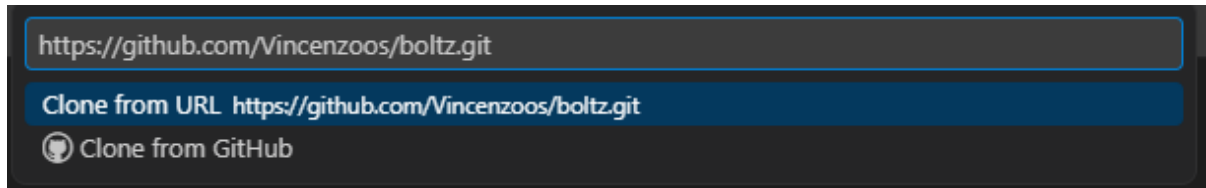


3. Clone the Boltz Repository

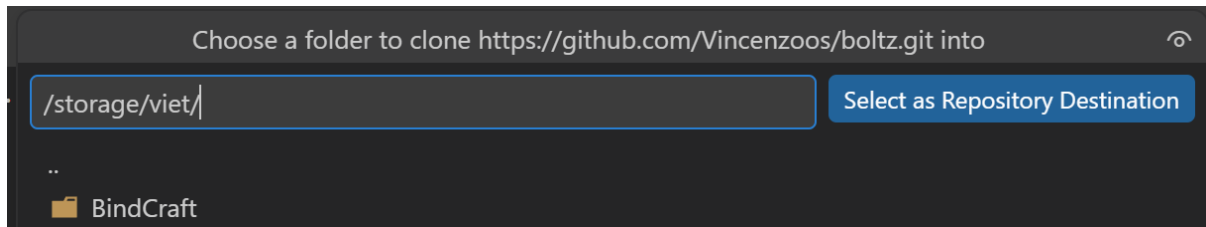
After connecting to Ewok, clone the Boltz repository from GitHub onto the remote machine.

Github repository: <https://github.com/Vincenzoos/boltz.git>

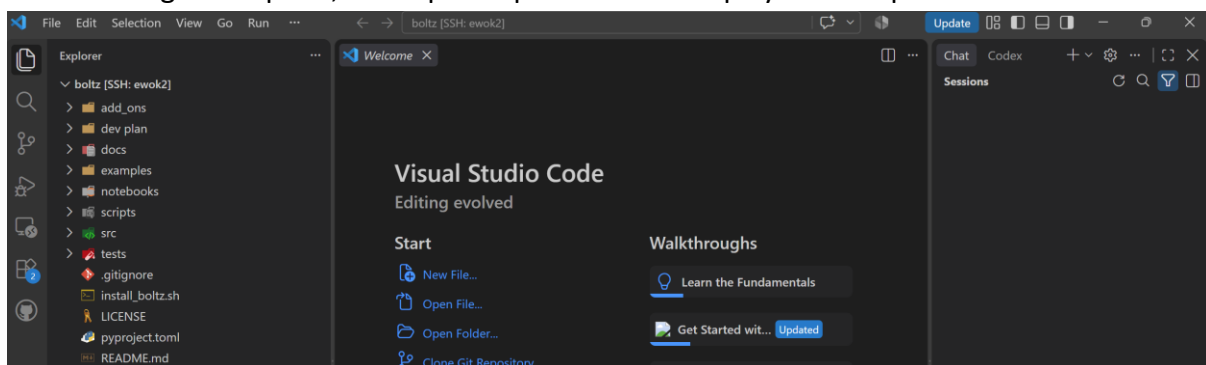
In the left sidebar, open Explorer, select Clone Repository, and enter the repository URL when prompted.



/home or /storage are two suitable locations to store the project. /storage is preferred if additional disk space needed.



Once cloning is complete, the Explorer panel should display the complete Boltz file structure.



4. Set Up the Boltz Environment

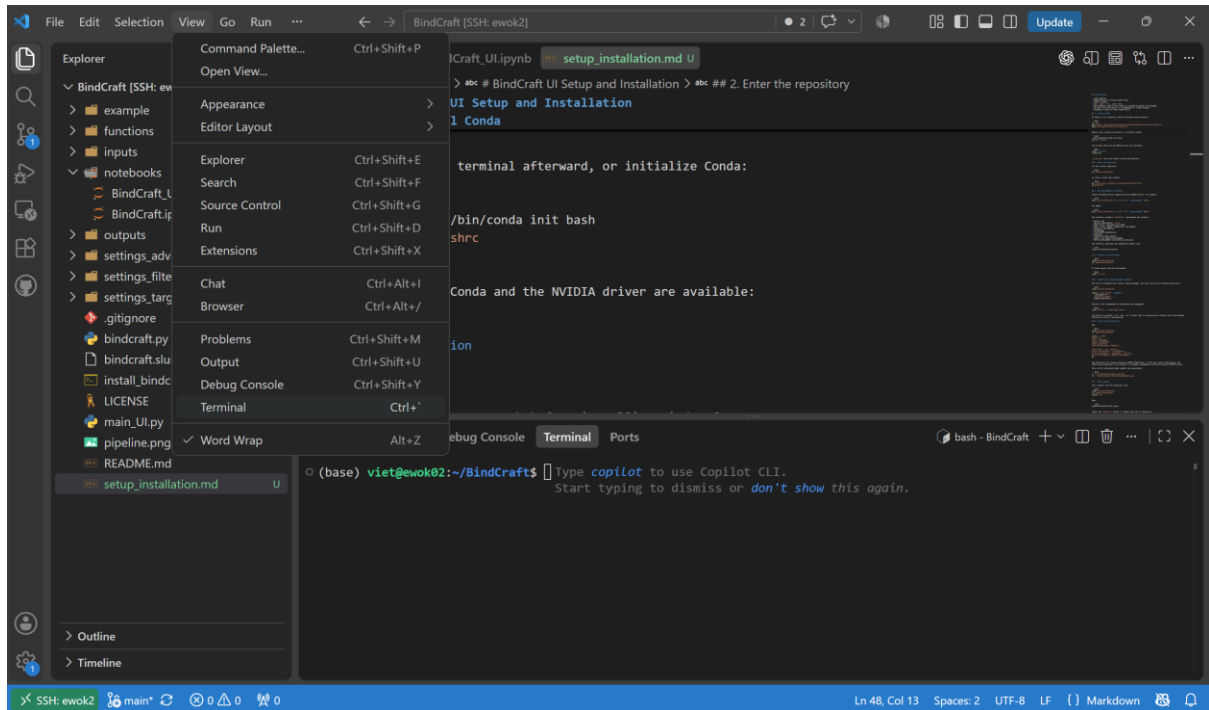
Before running the Boltz UI notebook, install the Boltz dependencies and the notebook UI dependencies.

4.1 Install Conda

If Conda is not installed on the Ewok machine, install Miniconda using the terminal:

```
cd /tmp
wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh
bash Miniconda3-latest-Linux-x86_64.sh
```

To access the terminal in VS Code, navigate to the bottom panel and expand it, or select View > Terminal.



Run the following commands to verify that Conda and the NVIDIA driver are available:

```
conda --version
nvidia-smi
```

Expected result:

```

(base) viet@ewok02:~/BindCraft$ conda --version
nvidia-smi
conda 26.5.3
Wed Aug 26 10:05:18 2026

+-----+
| NVIDIA-SMI 535.274.02                Driver Version: 535.274.02    CUDA Version: 12.2   |
+-----+-----+
| GPU Name                               Persistence-M   Bus-Id        Disp.A | Volatile Uncorr. ECC |
| Fan  Temp  Perf    Pwr:Usage/Cap       Memory-Usage   GPU-Util  Compute M. |
|                                           MIG M.         |
+-----+-----+
| 0  NVIDIA GeForce RTX 4080 ...        Off | 00000000:0B:00.0 Off |           N/A |
| 0%   47C   P8             9W / 320W           1MiB / 16376MiB |    0%      Default |
|                                           N/A           |
+-----+-----+
| 1  NVIDIA GeForce RTX 4080 ...        Off | 00000000:41:00.0 Off |           N/A |
| 0%   41C   P8            11W / 320W           1MiB / 16376MiB |    0%      Default |
|                                           N/A           |
+-----+-----+

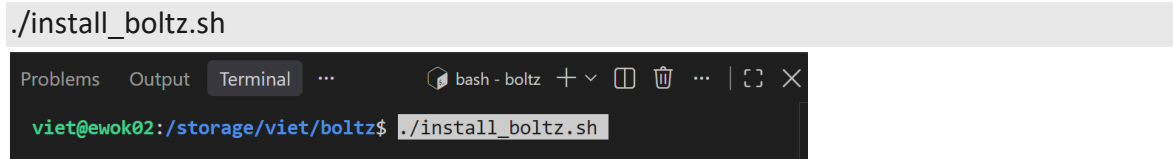
+-----+
| Processes:                               |
| GPU  GI  CI           PID   Type   Process name                        GPU Memory |
|   ID   ID                                     Usage      |
+-----+-----+
| No running processes found              |
+-----+

```

4.2 Run the Boltz Installer

Specify a CUDA version supported by the NVIDIA driver on the Ewok machine. For example:

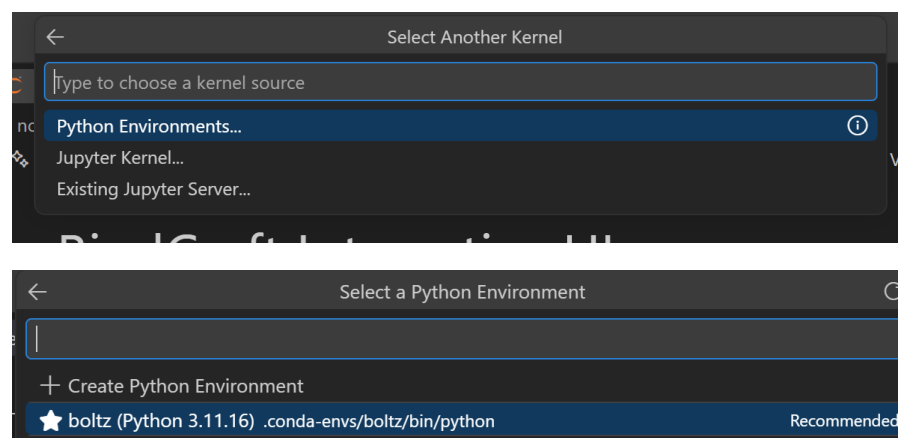
```
./install_boltz.sh
```



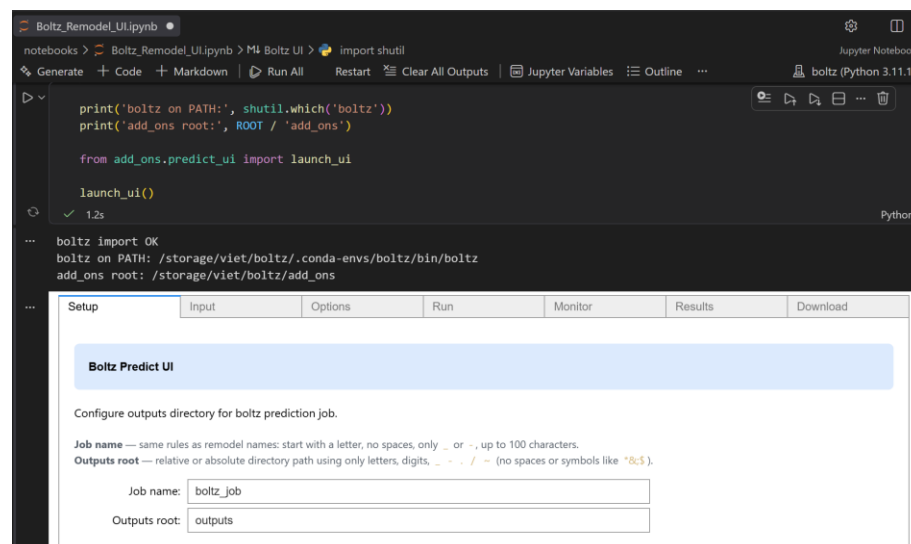
4.3 Run the Boltz UI Notebook

Open the Boltz_Remodel_UI.ipynb notebook in the notebooks folder.

Select the Boltz Conda environment as the notebook kernel:



Then click **Run All** for dependencies installation and UI rendering.



If the UI does not appear, you may need to reload the notebook to ensure that the ipywidgets dependency is loaded correctly.